EXPERIMENT NO. 10

End-to-End Data Analytics Project:

Business Intelligence and Predictive Analytics

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SECTION : B

ROLL NO. : 32

AIM

To implement a complete data analytics workflow on a real-world dataset by performing data preprocessing, exploratory data analysis,

statistical analysis, visualization, predictive modeling, and

presentation of actionable business insights.

STEP 1: IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import tkinter as tk

from tkinter import filedialog

from scipy.stats import pearsonr
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

STEP 2: SELECT DATASET

```
from google.colab import files

uploaded = files.upload()

if not uploaded:
    raise ValueError("No dataset selected.")

# Assuming only one file is uploaded, get its name
file_name = next(iter(uploaded))
df = pd.read_csv(file_name)
```

Choose Files Advertising ...d Sales.csv

Advertising Budget and Sales.csv(text/csv) - 4802 bytes, last modified: 12/25/2021 - 100% done

Saving Advertising Budget and Sales.csv to Advertising Budget and Sales.csv

STEP 3: DISPLAY BASIC DATASET INFORMATION

```
print("Dataset Shape:", df.shape)

print("\nDataset Columns:")
print(df.columns.tolist())

print("\nFirst Five Rows:")
print(df.head())
```

```
print("\nDataset Information:")
df.info()
```

```
Dataset Shape: (200, 5)

Dataset Columns:
['Unnamed: 0', 'TV Ad Budget ($)', 'Radio Ad Budget ($)', 'Newspaper Ad Budget ($)', 'Sales ($)']

First Five Rows:
   Unnamed: 0  TV Ad Budget ($)  Radio Ad Budget ($)  Newspaper Ad Budget ($)  \
0           1             230.1                 37.8                     69.2
1           2              44.5                 39.3                     45.1
2           3              17.2                 45.9                     69.3
3           4             151.5                 41.3                     58.5
4           5             180.8                 10.8                     58.4

   Sales ($)
0       22.1
1       10.4
2        9.3
3       18.5
4       12.9

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Unnamed: 0              200 non-null    int64
 1   TV Ad Budget ($)        200 non-null    float64
 2   Radio Ad Budget ($)     200 non-null    float64
 3   Newspaper Ad Budget ($)  200 non-null    float64
 4   Sales ($)               200 non-null    float64
dtypes: float64(4), int64(1)
memory usage: 7.9 KB
```

STEP 4: DATA PREPROCESSING

```
# Remove unnecessary index column if present
if "Unnamed: 0" in df.columns:
    df.drop("Unnamed: 0", axis=1, inplace=True)

# Remove duplicate rows
df.drop_duplicates(inplace=True)

# Convert required columns to numeric
numeric_columns = [
    "TV Ad Budget ($)",
    "Radio Ad Budget ($)",
    "Newspaper Ad Budget ($)",
    "Sales ($)"
]

for column in numeric_columns:
    df[column] = pd.to_numeric(
        df[column],
        errors="coerce"
    )

# Handle missing values
df[numeric_columns] = df[numeric_columns].fillna(
    df[numeric_columns].median()
)

print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())

print("\nDuplicate Rows After Preprocessing:", df.duplicated().sum())
```

```
Missing Values After Preprocessing:
TV Ad Budget ($)           0
Radio Ad Budget ($)        0
Newspaper Ad Budget ($)    0
Sales ($)                  0
dtype: int64

Duplicate Rows After Preprocessing: 0
```

STEP 5: DESCRIPTIVE STATISTICS

```
print("\nDescriptive Statistics:")
print(df.describe())
```

```
Descriptive Statistics:
       TV Ad Budget ($)  Radio Ad Budget ($)  Newspaper Ad Budget ($)  \
count        200.000000           200.000000               200.000000
mean         147.042500            23.264000                30.554000
std           85.854236            14.846809                21.778621
min            0.700000             0.000000                 0.300000
25%           74.375000             9.975000                12.750000
50%          149.750000            22.900000                25.750000
75%          218.825000            36.525000                45.100000
max          296.400000            49.600000               114.000000

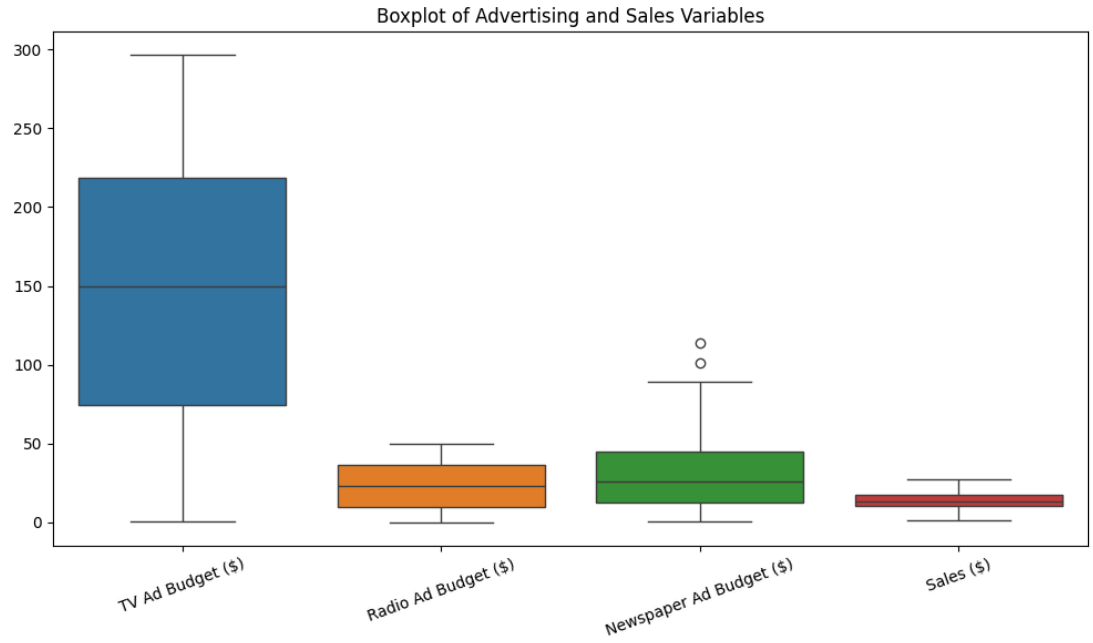
        Sales ($)
count  200.000000
mean    14.022500
std      5.217457
min      1.600000
25%     10.375000
50%     12.900000
75%     17.400000
max     27.000000
```

STEP 6: OUTLIER ANALYSIS

```
plt.figure(figsize=(10, 6))

sns.boxplot(data=df)

plt.title("Boxplot of Advertising and Sales Variables")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



STEP 7: DISTRIBUTION ANALYSIS

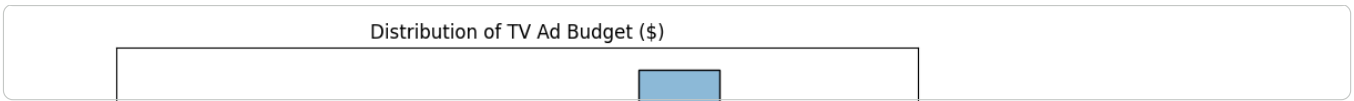
```
for column in numeric_columns:

    plt.figure(figsize=(8, 5))

    sns.histplot(
        df[column],
        kde=True
    )

    plt.title("Distribution of " + column)
    plt.xlabel(column)
    plt.ylabel("Frequency")

    plt.tight_layout()
    plt.show()
```



STEP 8: CORRELATION ANALYSIS

```
correlation_matrix = df[numeric_columns].corr()

print("\nCorrelation Matrix:")
print(correlation_matrix)

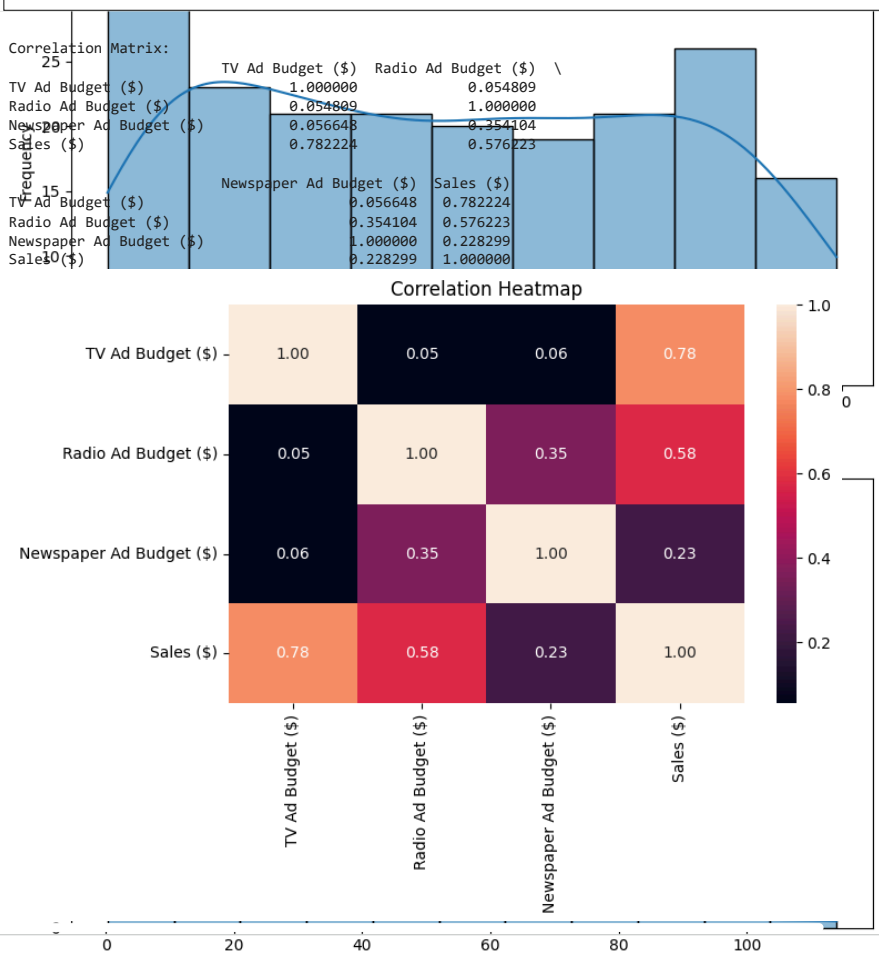
plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    fmt=".2f"
)

plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

```
Correlation Matrix:
                     TV Ad Budget ($)  Radio Ad Budget ($)  \
TV Ad Budget ($)             1.000000             0.054809
Radio Ad Budget ($)          0.054809             1.000000
Newspaper Ad Budget ($)      0.056648             0.354104
Sales ($)                    0.782224             0.576223

                     Newspaper Ad Budget ($)  Sales ($)
TV Ad Budget ($)                    0.056648   0.782224
Radio Ad Budget ($)                 0.354104   0.576223
Newspaper Ad Budget ($)             1.000000   0.228299
Sales ($)                           0.228299   1.000000
```



STEP 9: ADVERTISING BUDGET VS SALES

```
advertising_columns = [
    "TV Ad Budget ($)",
    "Radio Ad Budget ($)",
    "Newspaper Ad Budget ($)"
]

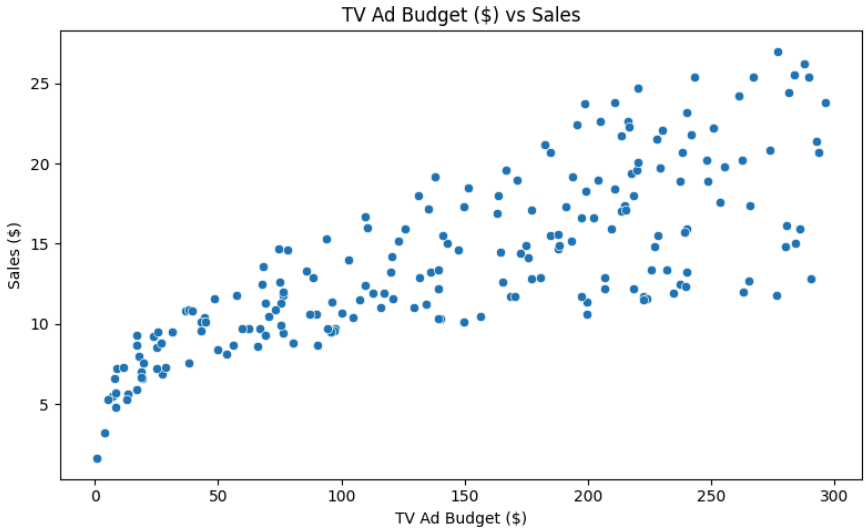
for column in advertising_columns:

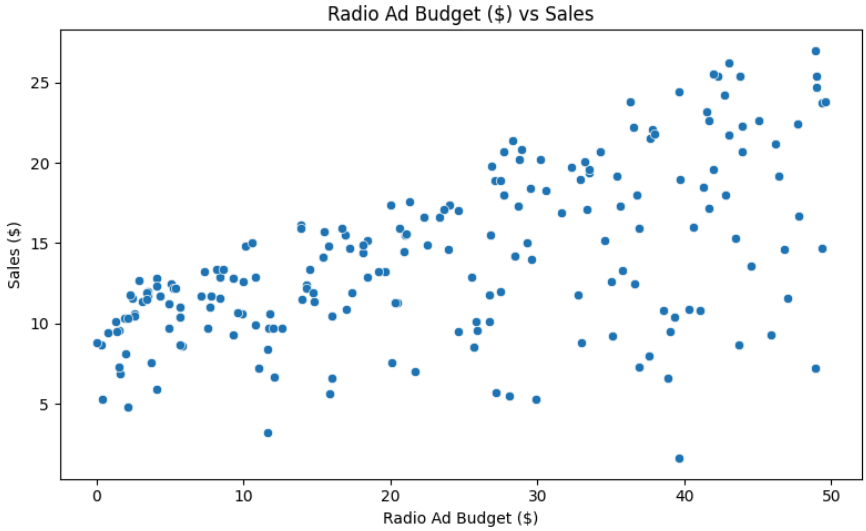
    plt.figure(figsize=(8, 5))

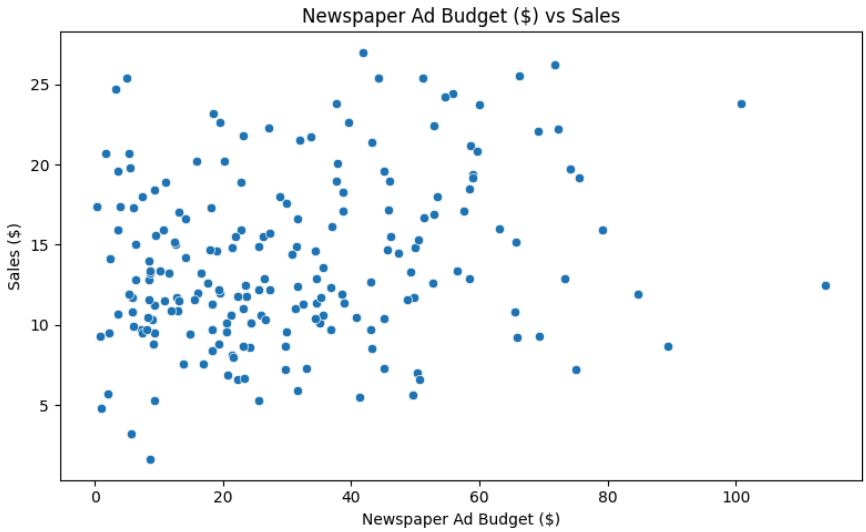
    sns.scatterplot(
        data=df,
        x=column,
        y="Sales ($)"
    )
```

```
    plt.title(column + " vs Sales")
    plt.xlabel(column)
    plt.ylabel("Sales ($)")

    plt.tight_layout()
    plt.show()
```







STEP 10: STATISTICAL ANALYSIS

```
print("\n========== PEARSON CORRELATION TEST ==========")

for column in advertising_columns:
```

```
    correlation, p_value = pearsonr(
        df[column],
        df["Sales ($)"]
    )

    print("\n", column)
    print("Correlation:", round(correlation, 4))
    print("P-value:", round(p_value, 6))
```

```
========== PEARSON CORRELATION TEST ==========

 TV Ad Budget ($)
Correlation: 0.7822
P-value: 0.0

 Radio Ad Budget ($)
Correlation: 0.5762
P-value: 0.0

 Newspaper Ad Budget ($)
Correlation: 0.2283
P-value: 0.001148
```

STEP 11: PREPARE DATA FOR MACHINE LEARNING

```
X = df[advertising_columns]
y = df["Sales ($)"]

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("\nTraining Samples:", X_train.shape[0])
print("Testing Samples:", X_test.shape[0])
```

```
Training Samples: 160
Testing Samples: 40
```

STEP 12: BUILD LINEAR REGRESSION MODEL

```
model = LinearRegression()

model.fit(
    X_train,
    y_train
)
```

```
▾ LinearRegression
LinearRegression()
```

STEP 13: MAKE PREDICTIONS

```
y_pred = model.predict(X_test)
```

STEP 14: MODEL EVALUATION

```
mae = mean_absolute_error(
    y_test,
    y_pred
)

mse = mean_squared_error(
    y_test,
    y_pred
)
```

```
rmse = np.sqrt(mse)

r2 = r2_score(
    y_test,
    y_pred
)

print("\n========== MODEL PERFORMANCE ==========")
print("Mean Absolute Error (MAE):", round(mae, 4))
print("Mean Squared Error (MSE):", round(mse, 4))
print("Root Mean Squared Error (RMSE):", round(rmse, 4))
print("R² Score:", round(r2, 4))
```

```
========== MODEL PERFORMANCE ==========
Mean Absolute Error (MAE): 1.4608
Mean Squared Error (MSE): 3.1741
Root Mean Squared Error (RMSE): 1.7816
R² Score: 0.8994
```

STEP 15: MODEL COEFFICIENTS

```
print("\n========== MODEL COEFFICIENTS ==========")

for column, coefficient in zip(
    advertising_columns,
    model.coef_
):
    print(column, ":", round(coefficient, 4))

print("\nIntercept:", round(model.intercept_, 4))
```

```
========== MODEL COEFFICIENTS ==========
TV Ad Budget ($) : 0.0447
Radio Ad Budget ($) : 0.1892
Newspaper Ad Budget ($) : 0.0028

Intercept: 2.9791
```

STEP 16: REGRESSION EQUATION

```
print("\n========== REGRESSION EQUATION ==========")

print(
    "Sales =",
    round(model.intercept_, 4),
    "+",
    round(model.coef_[0], 4),
    "* TV",
    "+",
    round(model.coef_[1], 4),
    "* Radio",
    "+",
    round(model.coef_[2], 4),
    "* Newspaper"
)
```

```
========== REGRESSION EQUATION ==========
Sales = 2.9791 + 0.0447 * TV + 0.1892 * Radio + 0.0028 * Newspaper
```

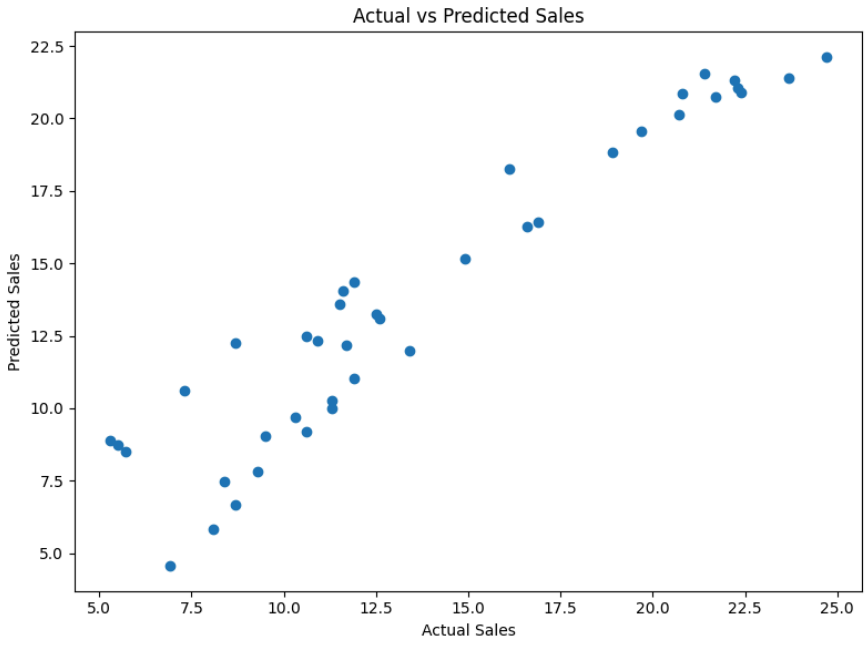
STEP 17: ACTUAL VS PREDICTED SALES

```
plt.figure(figsize=(8, 6))

plt.scatter(
    y_test,
    y_pred
)

plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title("Actual vs Predicted Sales")
```

```
plt.tight_layout()
plt.show()
```



STEP 18: RESIDUAL ANALYSIS

```
residuals = y_test - y_pred

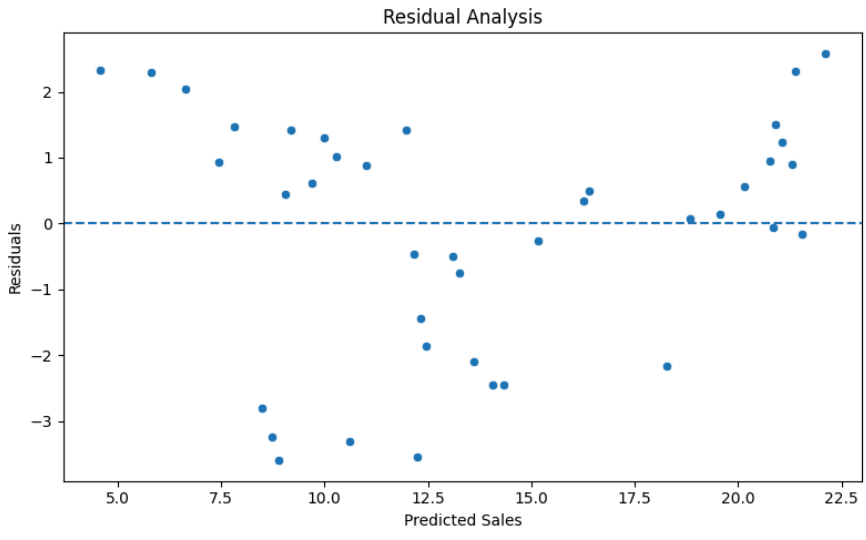
plt.figure(figsize=(8, 5))

sns.scatterplot(
    x=y_pred,
    y=residuals
)

plt.axhline(
    0,
    linestyle="--"
)

plt.xlabel("Predicted Sales")
plt.ylabel("Residuals")
plt.title("Residual Analysis")

plt.tight_layout()
plt.show()
```



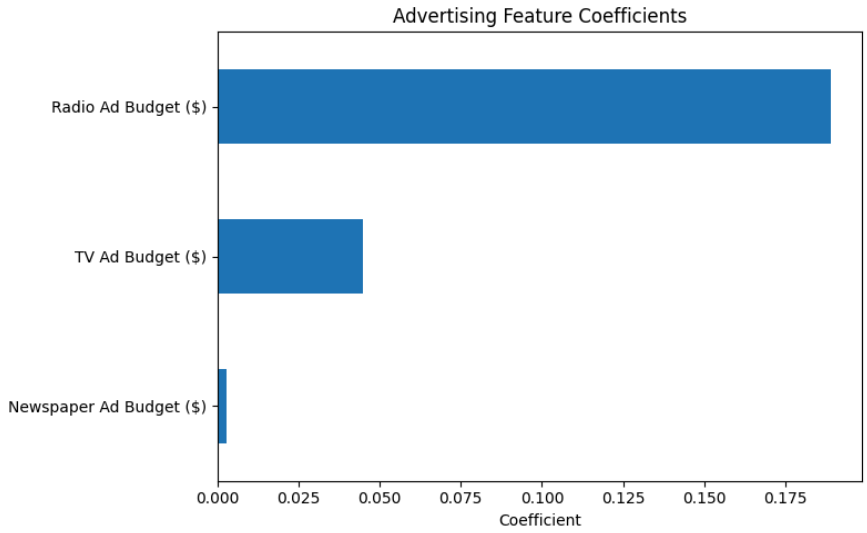
STEP 19: FEATURE IMPORTANCE

```
feature_importance = pd.Series(
    model.coef_,
    index=advertising_columns
)

plt.figure(figsize=(8, 5))

feature_importance.sort_values().plot(
    kind="barh"
)

plt.title("Advertising Feature Coefficients")
plt.xlabel("Coefficient")
plt.tight_layout()
plt.show()
```



STEP 20: BUSINESS INSIGHTS

```
print("\n========== BUSINESS INSIGHTS ==========")

strongest_feature = feature_importance.abs().idxmax()

print(
    "1. The advertising variable with the largest absolute "
    "regression coefficient is:",
    strongest_feature
)

print(
    "2. The model explains approximately",
    round(r2 * 100, 2),
    "% of the variation in sales."
)

print(
    "3. Advertising budgets can be analyzed to understand "
    "their relationship with sales."
)

print(
    "4. Actual and predicted sales can be compared to evaluate "
    "the usefulness of the predictive model."
)

print(
    "5. Correlation and regression analysis help identify "
    "important advertising factors."
)
```

```
========== BUSINESS INSIGHTS ==========
1. The advertising variable with the largest absolute regression coefficient is: Radio Ad Budget ($)
2. The model explains approximately 89.94 % of the variation in sales.
3. Advertising budgets can be analyzed to understand their relationship with sales.
```

```
4. Actual and predicted sales can be compared to evaluate the usefulness of the predictive model.
5. Correlation and regression analysis help identify important advertising factors.
```

STEP 21: BUSINESS RECOMMENDATIONS

```
print("\n========== BUSINESS RECOMMENDATIONS ==========")

print(
    "1. Use the regression model to estimate expected sales "
    "from advertising budgets."
)

print(
    "2. Monitor advertising channels that show stronger "
    "relationships with sales."
)

print(
    "3. Allocate advertising budgets using historical "
    "performance and predictive analysis."
)

print(
    "4. Periodically retrain the model using new sales data."
)

print(
    "5. Combine predictive analytics with dashboards for "
    "continuous business monitoring."
)
```

```
========== BUSINESS RECOMMENDATIONS ==========
1. Use the regression model to estimate expected sales from advertising budgets.
2. Monitor advertising channels that show stronger relationships with sales.
3. Allocate advertising budgets using historical performance and predictive analysis.
4. Periodically retrain the model using new sales data.
5. Combine predictive analytics with dashboards for continuous business monitoring.
```

QUESTIONS AND ANSWERS

Q1. What are the major stages involved in an end-to-end data analytics project?

Answer: The major stages are data collection, data preprocessing, exploratory data analysis, statistical analysis, visualization, predictive modeling, model evaluation, interpretation of results, and communication of business insights.

Q2. Why is data preprocessing considered the foundation of successful analytics and machine learning?

Answer: Data preprocessing improves data quality by handling missing values, duplicates, incorrect data types, and unnecessary variables. High-quality data helps analytical and machine learning models produce more reliable results.

Q3. How does Exploratory Data Analysis (EDA) contribute to model development?

Answer: EDA helps understand the structure and characteristics of data. It identifies patterns, relationships, outliers, and possible problems that can affect model development.

Q4. What factors should be considered while selecting a machine learning algorithm for a business problem?

Answer: The type of problem, size and structure of the dataset, target variable, number of features, interpretability requirements, computational resources, and required performance should be considered.

Q5. Why is model evaluation essential before deploying a predictive model?

Answer: Model evaluation determines how well a model performs on unseen data. It helps identify errors, overfitting, underfitting, and whether the model is suitable for practical use.

Q6. Explain how Business Intelligence tools complement machine learning in data analytics projects.

Answer: Business Intelligence tools provide dashboards, interactive charts, and reports for monitoring business performance. Machine learning provides predictive and analytical capabilities. Together they help organizations understand current performance and support future decision-making.

Q7. What challenges are commonly encountered while working with real-world datasets?

Answer: Common challenges include missing values, duplicate records, inconsistent data, outliers, incorrect data types, large datasets, categorical variables, and noisy information.

Q8. How can analytical insights be converted into actionable business recommendations?

Answer: Analytical findings should be connected with business objectives. Important trends, opportunities, risks, and performance gaps can be converted into specific actions such as budget allocation, resource planning, customer targeting, and process improvement.

Q9. Why is effective presentation and data storytelling important in analytics projects?

Answer: Effective presentation makes complex analytical findings easier to understand. Data storytelling connects results with business problems and helps stakeholders make informed decisions.

Q10. Suggest future improvements or advanced techniques that could enhance the accuracy and effectiveness of the developed analytics solution.

Answer: Future improvements may include collecting more recent data, feature engineering, cross-validation, hyperparameter tuning, regularization, ensemble models, advanced regression techniques, interactive dashboards, and periodic model retraining.